Building the atoms

Pauli principle

Two electrons cannot have identical quantum numbers $\{n, \ell, m_\ell, m_s\}$.

Very deep principle, related to indistinguishability of identical particles

- Fermions → spin halfinteger entities
  e.g. p, n obey Pauli principle
- Bosons → integer spin particles
  do not obey P.P.

e.g.:

$^3He = 2 \times p + n + 2 \cdot e^- \Rightarrow$ spin ½ fermion

$^4He = 2 \cdot p + 2 \cdot n + 2\, e^- \Rightarrow$ spin ∅ boson ...

electrons populate

shells K, L, M, N, ...

n = 1 n = 2 ...

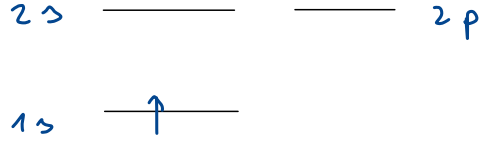
divided onto

subshells K → 1s, L → 2s, 2p, ...

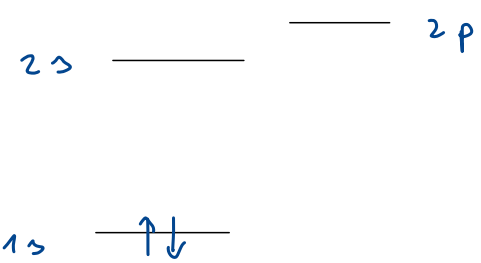
of electrons hosted:

shell	q. numbers	degeneracy
1s	$\ell = 0$ $m = 0$ $m_s = \pm \frac{1}{2}$	2
2s	$\ell = 0$ $m = 0$ $m_s = \pm \frac{1}{2}$	2
2p	$\ell = 1$ $m = 0, \pm 1$, $m_s = \pm \frac{1}{2}$	6
3s	$\ell = 0$ $m = 0$ $m_s = \pm \frac{1}{2}$	2
3p	$\ell = 1$ ⋮	6
3d	$\ell = 2$	10

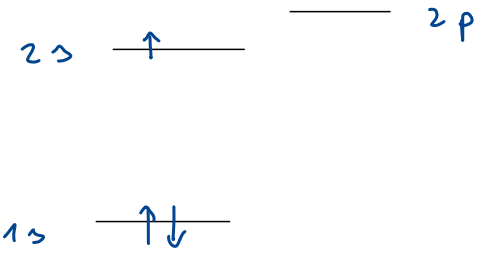
H $1s^1$



He $1s^2$



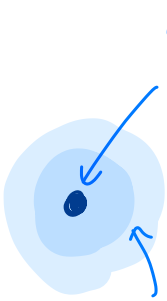
Li $1s^2\,2s^1$



Why is 2s lower than 2p ?

1s orbitals are close

2s orbitals more extended



electrons far away see charge $Q_T = e$!

$1s^2$ electrons with charge $Q_{1s} = -2e$

to see strong attractiv potential of Q_p one must get close to origin.

But

$P_{2s} \sim e^{-r/a_0}$ ← finite at $r=0$

$P_{2p} \sim r^2 e^{-r/a_0}$ ← vanishes at $r=0$

"s" orbitals see also the charge at $r=0$!

"p" orbitals cannot get close

$\Rightarrow \quad E_{2s} < E_{2p}$

$E_{3s} < E_{3p} < E_{3d}$

$\vdots$

Within a shell, subshells are filled in this order $s \to p \to d \to f \ldots$

Order of main shells is, however, scrambled:

$1s \to 2s, 2p \to 3s, 3p \to 4s \to 3d \to 4p$

$1s$: H, He

$2s, 2p$: Li, Be, B, C, N, O, F, Ne

Li: Alkalis; Be: Alkaline earth elements — reactive, loose e^-

F: Halogens — valence -1 active

Ne: Noble gases — closed s,p shells

$3s, 3p$: Na, ..., Ar

$4s$: K, Ca intruder ...

$3d$: Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn

transition metals ~ ferro-magnets

$4s \to$ metal

$3d \to$ ferromagnet !

$4p$: Ga, Ge, As, Se, Br, Kr

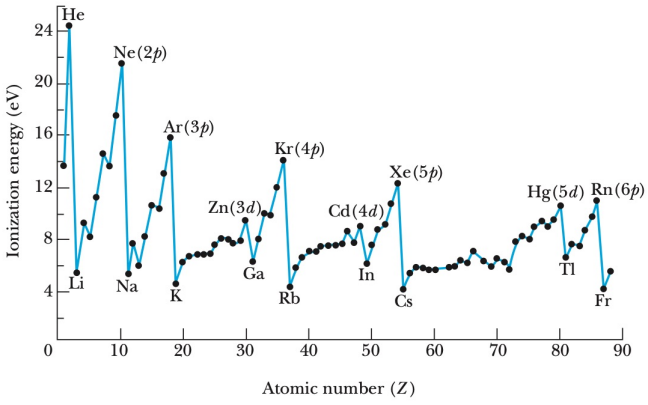
$5s$, $4d$ } mix

Periodic table

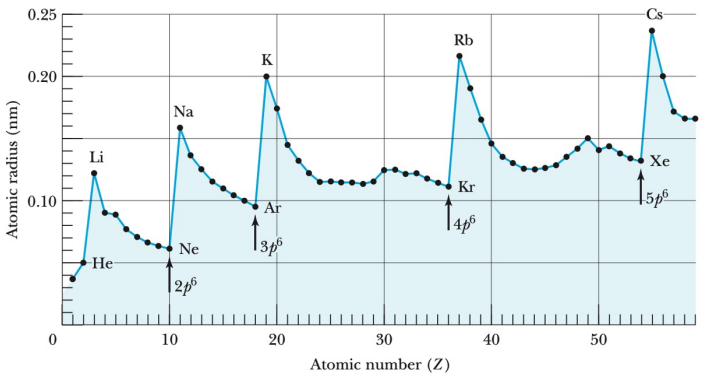
Groups:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
			Transition elements															
	1 H $1s$																	2 He $1s^2$
$1s^2$	3 Li $2s^1$	4 Be $2s^2$											5 B $2s^2\,2p^1$	6 C $2s^2\,2p^2$	7 N $2s^2\,2p^3$	8 O $2s^2\,2p^4$	9 F $2s^2\,2p^5$	10 Ne $2s^2\,2p^6$
$2s^2 2p^6$	11 Na $3s^1$	12 Mg $3s^2$											13 Al $3s^2\,3p^1$	14 Si $3s^2\,3p^2$	15 P $3s^2\,3p^3$	16 S $3s^2\,3p^4$	17 Cl $3s^2\,3p^5$	18 Ar $3s^2\,3p^6$
$3s^2 3p^6$	19 K $4s^1$	20 Ca $4s^2$	21 Sc $3d^1\,4s^2$	22 Ti $3d^2\,4s^2$	23 V $3d^3\,4s^2$	24 Cr $3d^5\,4s^1$	25 Mn $3d^5\,4s^2$	26 Fe $3d^6\,4s^2$	27 Co $3d^7\,4s^2$	28 Ni $3d^8\,4s^2$	29 Cu $3d^{10}\,4s^1$	30 Zn $3d^{10}\,4s^2$	31 Ga $3d^{10}\,4s^2\,4p^1$	32 Ge $3d^{10}\,4s^2\,4p^2$	33 As $3d^{10}\,4s^2\,4p^3$	34 Se $3d^{10}\,4s^2\,4p^4$	35 Br $3d^{10}\,4s^2\,4p^5$	36 Kr $3d^{10}\,4s^2\,4p^6$
$3d^{10}4s^2 4p^6$	37 Rb $5s^1$	38 Sr $5s^2$	39 Y $4d^1\,5s^2$	40 Zr $4d^2\,5s^2$	41 Nb $4d^4\,5s^1$	42 Mo $4d^5\,5s^1$	43 Tc $4d^6\,5s^1$	44 Ru $4d^7\,5s^1$	45 Rh $4d^8\,5s^1$	46 Pd $4d^{10}$	47 Ag $4d^{10}\,5s^1$	48 Cd $4d^{10}\,5s^2$	49 In $4d^{10}\,5s^2\,5p^1$	50 Sn $4d^{10}\,5s^2\,5p^2$	51 Sb $4d^{10}\,5s^2\,5p^3$	52 Te $4d^{10}\,5s^2\,5p^4$	53 I $4d^{10}\,5s^2\,5p^5$	54 Xe $4d^{10}\,5s^2\,5p^6$
$4d^{10}5s^2 5p^6$	55 Cs $6s^1$	56 Ba $6s^2$	57 La $5d^1\,6s^2$	72 Hf $4f^{14}\,5d^2\,6s^2$	73 Ta $4f^{14}\,5d^3\,6s^2$	74 W $4f^{14}\,5d^4\,6s^2$	75 Re $4f^{14}\,5d^5\,6s^2$	76 Os $4f^{14}\,5d^6\,6s^2$	77 Ir $4f^{14}\,5d^9$	78 Pt $4f^{14}\,5d^9\,6s^1$	79 Au $4f^{14}\,5d^{10}\,6s^1$	80 Hg $4f^{14}\,5d^{10}\,6s^2$	81 Tl $4f^{14}\,5d^{10}\,6s^2\,6p^1$	82 Pb $4f^{14}\,5d^{10}\,6s^2\,6p^2$	83 Bi $4f^{14}\,5d^{10}\,6s^2\,6p^3$	84 Po $4f^{14}\,5d^{10}\,6s^2\,6p^4$	85 At $4f^{14}\,5d^{10}\,6s^2\,6p^5$	86 Rn $4f^{14}\,5d^{10}\,6s^2\,6p^6$
$4f^{14}5d^{10}6s^2 6p^6$	87 Fr $7s^1$	88 Ra $7s^2$	89 Ac $6d^1\,7s^2$	104 Rf $5f^{14}\,6d^2\,7s^2$	105 Db $5f^{14}\,6d^3\,7s^2$	106 Sg $5f^{14}\,6d^4\,7s^2$	107 Bh $5f^{14}\,6d^5\,7s^2$	108 Hs $5f^{14}\,6d^6\,7s^2$	109 Mt $5f^{14}\,6d^7\,7s^2$	110 Ds $5f^{14}\,6d^9\,7s^1$	111 Rg $5f^{14}\,6d^{10}\,7s^1$	112 Cn $5f^{14}\,6d^{10}\,7s^2$	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

	58	59	60	61	62	63	64	65	66	67	68	69	70	71
Lanthanides	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	$4f^2\,6s^2$	$4f^3\,6s^2$	$4f^4\,6s^2$	$4f^5\,6s^2$	$4f^6\,6s^2$	$4f^7\,6s^2$	$4f^7\,6s^2\,5d^1$	$4f^9\,6s^2$	$4f^{10}\,6s^2$	$4f^{11}\,6s^2$	$4f^{12}\,6s^2$	$4f^{13}\,6s^2$	$4f^{14}\,6s^2$	$4f^{14}\,5d^1\,6s^2$
	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Actinides	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
	$6d^2\,7s^2$	$5f^2\,6d^1\,7s^2$	$5f^3\,6d^1\,7s^2$	$5f^4\,6d^1\,7s^2$	$5f^6\,7s^2$	$5f^7\,7s^2$	$5f^7\,6d^1\,7s^2$	$5f^8\,6d^1\,7s^2$	$5f^{10}\,7s^2$	$5f^{11}\,7s^2$	$5f^{12}\,7s^2$	$5f^{13}\,7s^2$	$5f^{14}\,7s^2$	$5f^{14}\,6d^1\,7s^2$

Ionization energies

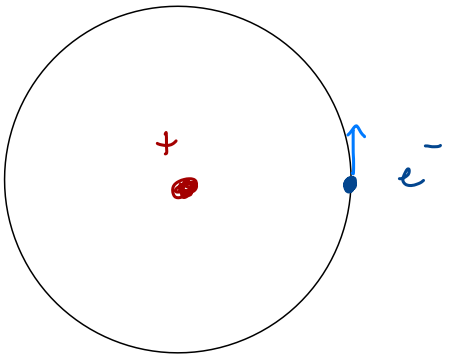


Atomic radii

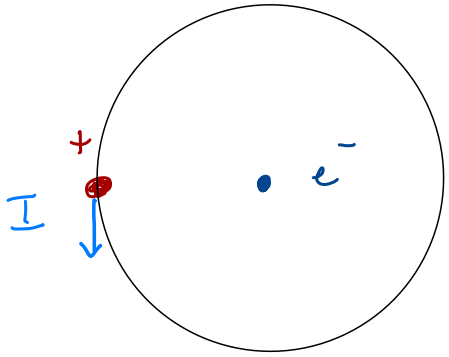


L S coupling and fine structure

Consider Hydrogen



in e^-'s coordinate system



proton creates effective magnetic field at e^-

$$\underline{B}_{eff} \sim \underline{L}$$

$\Rightarrow$ energy of spin dipole in this field

$$E_{spin} = -\underline{B}_{eff} \cdot \underline{\mu}_s \sim \underline{S} \cdot \underline{L}$$

$$\underline{\mu}_s = -\frac{e}{2m} \cdot g\, \underline{S}$$

total angular momentum

$$\underline{J} = \underline{L} + \underline{S}$$ ← good quantum number!

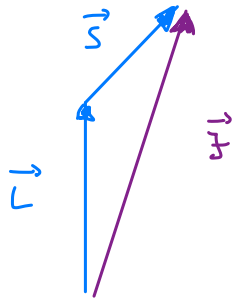
or in units of $\hbar$: $\underline{j} = \underline{\ell} + \underline{s}$

z-component: $j^z = m_\ell + m_s \Rightarrow$

m_ℓ: $-\ell, \dots, \ell$ $\quad$ m_s: $\pm \frac{1}{2}$

half-integers! $\quad$ Q: $j = ??$

$$\vec{j} = \vec{l} + \vec{s}$$



$$|\vec{j}_{max}| = |\vec{l}| + |\vec{s}| \quad \Rightarrow \quad j_{max} = l + s = l + \tfrac{1}{2}$$

$$|j_{min}| = \begin{cases} |\vec{l}| - |\vec{s}| & \text{if } |\vec{l}| > |\vec{s}| \\ |\vec{s}| - |\vec{l}| & \text{if } |\vec{s}| > |\vec{l}| \end{cases}$$

$$\Rightarrow \quad j_{min} = l - s = l - \tfrac{1}{2} \quad \text{if } l \neq 0$$

Example:

$l = 1$ →
- $j = \frac{3}{2}$, $m_j \in \{-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\}$
- $j = \frac{1}{2}$, $m_j \in \{-\frac{1}{2}, \frac{1}{2}\}$

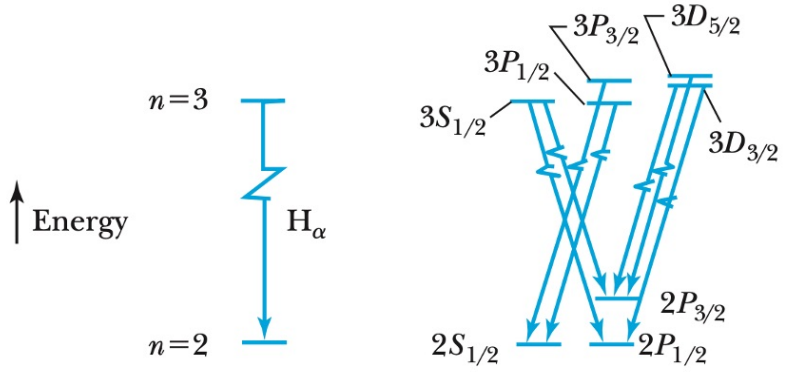
total number of states $2 + 4 = 6$ ✓

antiparallel alignment has lower energy due to $\underline{L} \cdot \underline{s}$ coupling!

2p ——— $j = 3/2$
——— $j = 1/2$ ← fine structure splitting $\sim 10^{-4}$ eV

1s ——— $j = 1/2$

Hydrogen terms:



Unperturbed (a) Fine structure (b)

notation: $n L_j$

transitions: $\Delta L = \pm 1$, $\Delta j = \pm 1, 0$

$\Delta m_j = \pm 1, 0$

Case of several electrons

Hund's rules:

I Electrons tend to align spins.
~ cannot get close (fermions)

II In a shell, they maximize L

III below half-filled shell → minimize J
~ spin-orbit coupling ($\underline{L} \cdot \underline{S}$)

consider 2-electron states:

$$\underline{S} = \underline{S}_1 + \underline{S}_2$$

$$\underline{L} = \underline{L}_1 + \underline{L}_2$$

$$\underline{J} = \underline{J}_1 + \underline{J}_2$$

total spin ($\hbar$ units) S
orbital momentum L
total angular momentum J

$$S = \frac{1}{2} + \frac{1}{2} = 1 \quad \text{or} \quad \frac{1}{2} - \frac{1}{2} = 0$$

$S = 1$: triplet; $S = 0$: singlet

L depends on shell

$\underline{\ell_1 = \ell_2 = 0}$ $\Rightarrow$ 2s $\uparrow\downarrow$ $\quad L = 0,\ S = 0,\ J = 0$

state $^{2S+1}L_J$ $\Rightarrow$ $^1S_0$ state

$\underline{\ell_1 = \ell_2 = 1}$

2p — $\uparrow$ $\uparrow$
$m = -1$ 0 1

Hund 1 $S = 1$ lowest energy state

Hund 2 $L_z^{max} = 1 + 0 = 1 \Rightarrow L = 1$

Hund 3 $J = L - S = 0$

$\Rightarrow$ $^3P_0$ lowest energy states

Remark: Pauli principle forbids certain states...

$S = 1 \rightarrow L = 2$ is NOT possible ($L = 1$ is)

$S = 0 \Rightarrow L = 2$ is ! possible !

$\uparrow\uparrow$ $\quad$ $\uparrow\downarrow$ $\quad$ ~~$\uparrow\uparrow$~~ $\quad$ $L_z = 2$

$m = 1$

$S = 0 \Rightarrow L = \text{even}$ $\quad$ $L = 2$ or $\Rightarrow J = 2$

$L = 0 \Rightarrow J = 0$

$S = 1 \Rightarrow L = \text{odd}$ $\quad$ $L = 1 \Rightarrow J = 2, 1, 0$

terms $\quad S = 0$

——— $^1D_2$

——— $^1S_0$

electron-electron interaction (exchange) $\updownarrow \sim$ eV

$S = 1$

——— $^3P_2$ $\quad \updownarrow$ fine structure (LS splitting) $\sim 10^{-4}$ eV

——— $^3P_1$

——— $^3P_0$

For heavy atoms LS coupling becomes large!

Selection rules: $\quad \Delta L = \pm 1 \quad \Delta S = 0$

$\Delta J = \pm 1, 0$ except

$J = 0 \rightarrow 0$ forbidden

Anomalous Zeeman splitting

$$\hat{H}_z = -\mu_B \vec{B}\left(\vec{L} + 2\vec{S}\right) = -$$

($2 = 1$ ↙)

B small << LS coupling $\Rightarrow$ $J = \text{cst}$

$$H_{eff} = -\mu_B B \cdot g\, m_J$$

(g: Landé factor)

$$g = g(J, S, L) = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2 \cdot J(J+1)}$$

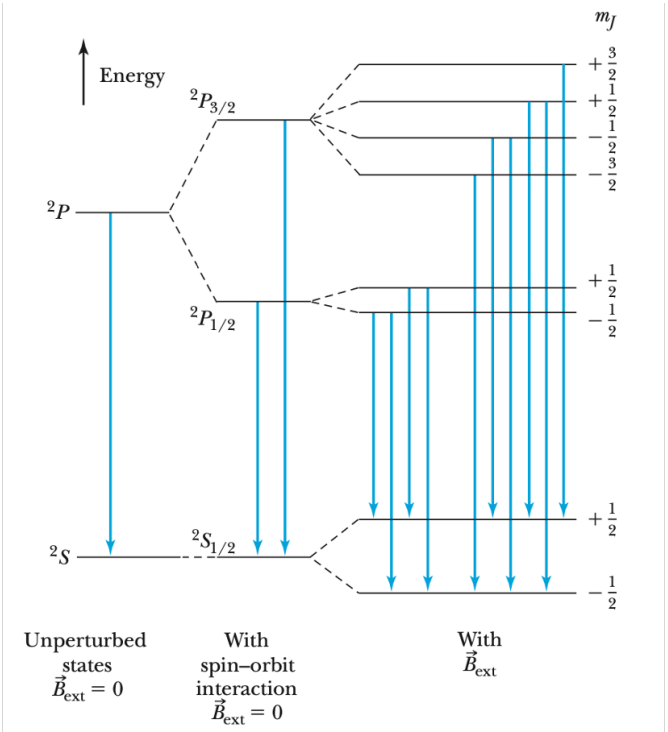
$S = 0 \Rightarrow J = L \Rightarrow g = 1$

$L = 0 \Rightarrow J = S \Rightarrow g = 2$

excited states of Na

$^2P_{1/2}$ $\quad S = J = 1/2 \quad L = 1 \Rightarrow g = 2 - \frac{2}{3/2} = 0,66$

$^2P_{3/2}$ $\quad S = 1/2 \quad J = 3/2 \quad L = 1 \Rightarrow g = 1,3$



$g = 1,33$

$g = 0,66$

$g = 2$